

QUOTES RECOMMENDATION CHATBOT USING NLP

Epic 2: ENVIRONMENT SETUP

Story 1: Setting Up the Rasa Project

1. Introduction

After successfully installing Rasa, the next step is to initialize the chatbot project structure. Rasa provides a built-in command that automatically generates a standard project template.

This structured template helps organize training data, configuration files, custom actions, and models in a systematic way, making development easier and more efficient.

2. Initializing the Rasa Project

To create the project structure, run the following command inside the activated virtual environment: (rasa init)

This command:

- Creates the required folders and configuration files
- Generates sample training data
- Trains an initial chatbot model
- Allows testing the chatbot immediately

Successful execution confirms that the Rasa project has been properly initialized.

3. Project Structure Created After Initialization

After running rasa init, the following directories and files are generated:

3.1 data/

This directory contains training data files:

- **nlu.yml**
 - Contains training examples for intent classification
 - Defines user messages for different intents

- **stories.yml**
 - Defines conversation flows
 - Trains the dialogue management system
 - **rules.yml**
 - Contains rule-based conversations
 - Handles fixed responses such as greetings and fallback cases
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3.2 actions/

- Used to define custom backend logic
 - Contains actions.py file
 - Allows integration with external APIs or databases
 - In this project, it supports web-based communication (e.g., Flask integration)
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3.3 models/

- Stores trained chatbot models
- Models are generated after running:

rasa train

- These models are used during chatbot execution
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3.4 tests/

- Contains test stories
 - Used for validating chatbot behavior
 - Helps ensure the chatbot responds correctly to user inputs
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3.5 config.yml

- Defines the NLU pipeline
- Specifies machine learning components

- Configures dialogue policies
 - Controls how user input is processed and how responses are selected
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3.6 domain.yml

- Acts as the chatbot's knowledge base
 - Defines:
 - Intents
 - Responses
 - Actions
 - Entities
 - Session configuration
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3.7 credentials.yml

- Configures communication channels
 - Enables REST API support
 - Allows web-based interaction with the chatbot
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3.8 endpoints.yml

- Defines service endpoints
 - Configures action server details
 - Enables connection between Rasa Core and custom actions
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4. Conclusion

Setting up the Rasa project using `rasa init` establishes a well-organized and ready-to-use chatbot development structure.

This step ensures:

- Proper file organization
- Smooth training and testing process

- Easy integration with web interfaces
- Scalable and maintainable chatbot architecture

With the project structure ready, the next phase involves customizing intents, training data, and dialogue flows for the Quotes Recommendation Chatbot.